The bound $A(32, 16, 13) \le 26$

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Notation. $A(n, d, w)$ is the maximum size of a binary code of length n, constant weight w and minimum distance at least d. A word of length 32 and weight 13 is regarded as a 13-subset of a set of 32 points (the coordinate positions). Two words c, c' are at distance $2(13 - |c \cap c'|)$, so their distance is at least 16 exactly when $|c \cap c'| \le 5$.

Theorem. $A(32, 16, 13) \le 26$.

The bounds known before were $25 \le A(32, 16, 13) \le 27$, see [3]. The Theorem does not determine $A(32, 16, 13)$.

Outline. Sections 1–4 are proved by hand. They show that a code with 27 words would have one of 102 possible "defect graphs", and that for each of them certain systems of linear equations would have a solution in nonnegative numbers. Section 5 states what is then checked by computer, in exact arithmetic: in each of the 102 cases a necessary condition fails. The number 102 of graphs is also confirmed there. Section 6 treats one of the cases by hand, as an illustration. No result on the completion of partial Hadamard matrices is used.

1 Counting in columns

Let $c_1, \dots, c_M$ be the words of a code. For $a \ne b$ put

$$\delta_{ab} = 5 - |c_a \cap c_b| \ \ge 0,$$

and for a point j let r_j be the number of words that contain j. Counting the pairs (word, point on it), and the triples (two words, common point), gives

$$\sum_j r_j = 13M, \qquad \sum_j \binom{r_j}{2} = \sum_{a<b} |c_a \cap c_b| = 5\binom{M}{2} - \sum_{a<b} \delta_{ab}. \tag{1}$$

For integers r and q,

$$\binom{r}{2} = qr - \frac{q(q+1)}{2} + \frac{(r-q)(r-q-1)}{2}, \tag{2}$$

and the last term is nonnegative, being half the product of two consecutive integers.

Lemma 1. *$M \le 27$. If $M = 27$, then*

$$\sum_{a<b} \delta_{ab} + \sum_j \frac{(r_j - 10)(r_j - 11)}{2} = 5. \tag{3}$$

Proof. Let $M = 28$. Adding (2) with $q = 11$ over the 32 points and using (1) gives $\sum_j \binom{r_j}{2} \ge 11 \cdot 13 \cdot 28 - 32 \cdot 66 = 1892$, while (1) gives $\sum_j \binom{r_j}{2} \le 5\binom{28}{2} = 1890$. So there is no code with 28 words, and hence none with more. Let $M = 27$. Adding (2) with $q = 10$ gives $\sum_j \binom{r_j}{2} = 10 \cdot 13 \cdot 27 - 32 \cdot 55 + \sum_j (r_j - 10)(r_j - 11)/2 = 1750 + \sum_j (r_j - 10)(r_j - 11)/2$, and by (1) the left side equals $5\binom{27}{2} - \sum_{a<b} \delta_{ab} = 1755 - \sum_{a<b} \delta_{ab}$. □

From now on $M = 27$. All terms in (3) are nonnegative integers. The *defect graph* G has the 27 words as vertices; a and b are joined by an edge of weight δ_{ab} whenever $\delta_{ab} > 0$. Let $D = \sum_{a<b} \delta_{ab}$ be its total weight. By (3), $D \le 5$, so G has at most 10 non-isolated vertices, and every point j satisfies

$$\frac{(r_j - 10)(r_j - 11)}{2} \le 5 - D. \tag{4}$$

A graph with edge weights of total D is the same thing as a multigraph without loops with D edges. Up to isomorphism, and disregarding isolated vertices, the number of such multigraphs with $D = 0, 1, 2, 3, 4, 5$ edges is $1, 1, 3, 8, 23, 66$, which is 102 in all. We show that none of these 102 graphs is the defect graph of a code. Fix one of them, G, with v non-isolated vertices, and let $u = 27 - v$.

2 The Gram matrix

Let X be the 27×32 matrix with entry -1 in row a and column j if $j \in c_a$, and $+1$ otherwise. The inner product of two rows is 32 minus twice their distance, that is, $-4\delta_{ab}$ for $a \ne b$, and 32 for $a = b$. Each row sum is $32 - 26 = 6$. So

$$XX^{\mathsf{T}} = H := 32\, I - 4\Delta, \qquad X\mathbf{1} = 6 \cdot \mathbf{1}, \tag{5}$$

where $\Delta = (\delta_{ab})$ has zero diagonal and $\mathbf{1}$ denotes an all-one vector (of length 32 on the left, 27 on the right). The matrix H depends only on G.

Lemma 2. *H is positive definite.*

Proof. Let $d_a = \sum_b \delta_{ab} \le D \le 5$. For a real vector y, using $2y_a y_b \le y_a^2 + y_b^2$,

$$y^{\mathsf{T}} H y = 32 \sum_a y_a^2 - 8 \sum_{a<b} \delta_{ab}\, y_a y_b \;\ge\; \sum_a (32 - 4d_a)\, y_a^2 \;\ge\; 12 \sum_a y_a^2. \qquad \square$$

Put

$$\beta = 6\, H^{-1}\mathbf{1}, \qquad S = 32 - 36 \cdot \mathbf{1}^{\mathsf{T}} H^{-1} \mathbf{1}.$$

For a point j let $z_j \in \{\pm 1\}^{27}$ be the j-th column of X.

Lemma 3. (i) $S \ge 0$.

(ii) *If $S > 0$, let P be the 32×32 matrix with entries*

$$P_{jj'} = z_j^{\mathsf{T}} H^{-1} z_{j'} + \frac{(1 - \beta^{\mathsf{T}} z_j)(1 - \beta^{\mathsf{T}} z_{j'})}{S}.$$

Then $I - P$ is positive semidefinite. In particular $P_{jj} \le 1$ for every point j.

(iii) *If $S = 0$, then $\beta^{\mathsf{T}} z_j = 1$ and $z_j^{\mathsf{T}} H^{-1} z_j \le 1$ for every point j.*

Proof. Let $W \subseteq \mathbb{R}^{32}$ be the row space of X. By Lemma 2 the orthogonal projection onto W is $\Pi = X^{\mathsf{T}} H^{-1} X$. By (5), $\Pi\mathbf{1} = X^{\mathsf{T}} H^{-1} \cdot 6 \cdot \mathbf{1} = X^{\mathsf{T}}\beta$, and $|\Pi\mathbf{1}|^2 = \mathbf{1}^{\mathsf{T}}\Pi\mathbf{1} = 36 \cdot \mathbf{1}^{\mathsf{T}} H^{-1}\mathbf{1}$. So the vector $\mathbf{1} - X^{\mathsf{T}}\beta$ is orthogonal to W and has squared length $32 - 36 \cdot \mathbf{1}^{\mathsf{T}} H^{-1}\mathbf{1} = S$. This proves (i). Its j-th entry is $1 - \beta^{\mathsf{T}} z_j$, and the (j, j') entry of Π is $z_j^{\mathsf{T}} H^{-1} z_{j'}$. If $S > 0$, then $P = \Pi + (\mathbf{1} - X^{\mathsf{T}}\beta)(\mathbf{1} - X^{\mathsf{T}}\beta)^{\mathsf{T}}/S$ is the orthogonal projection onto the space spanned by W and $\mathbf{1}$, so $I - P$ is positive semidefinite. If $S = 0$, then $\mathbf{1} = X^{\mathsf{T}}\beta$, and $I - \Pi$ is positive semidefinite. $\square$

3 Types of points

Number the words so that $c_1, \dots, c_v$ are the non-isolated vertices of G. Then H is the direct sum of the $v \times v$ matrix H_v belonging to these words and $32\, I_u$. Consequently $\beta = (\beta_v, \frac{3}{16}\mathbf{1})$ with $\beta_v = 6H_v^{-1}\mathbf{1}$.

The *type* of a point j is the pair (b, a), where $b \in \{0,1\}^v$ has $b_k = 1$ exactly when $j \in c_k$, and a is the number of the words $c_{v+1}, \dots, c_{27}$ that contain j. So $r_j = |b| + a$, where $|b| = \sum_k b_k$. Put $z_b = \mathbf{1} - 2b \in \{\pm 1\}^v$. For a point of type (b, a),

$$z_j^{\mathsf{T}} H^{-1} z_j = z_b^{\mathsf{T}} H_v^{-1} z_b + \frac{u}{32}, \qquad \beta^{\mathsf{T}} z_j = \beta_v^{\mathsf{T}} z_b + \frac{3}{16}(u - 2a). \tag{6}$$

So whether a point satisfies (4) and the conditions of Lemma 3 for a single point depends only on its type. A type (b, a), with $0 \le a \le u$, is called *admissible* if it satisfies (4) and, when $S > 0$, the condition $P_{jj} \le 1$; when $S = 0$, the conditions $\beta^{\mathsf{T}} z_j = 1$ and $z_j^{\mathsf{T}} H^{-1} z_j \le 1$. There are $2^v(u+1)$ types. If $v = 0$, the terms with H_v, β_v and z_b are absent.

For an admissible type $i = (b, a)$ let N_i be the number of points of type i. Then $N_i \ge 0$, and

(E1)	$\sum_i N_i = 32$,	
(E2)	$\sum_i a\, N_i = 13u$,	
(E3)	$\sum_i \binom{a}{2} N_i = 5\binom{u}{2}$,	
(E4)	$\sum_i b_k\, N_i = 13$	$(1 \le k \le v)$,
(E5)	$\sum_i a\, b_k\, N_i = 5u$	$(1 \le k \le v)$,
(E6)	$\sum_i b_k b_l\, N_i = 5 - \delta_{kl}$	$(1 \le k < l \le v)$.

Indeed, these count: the points; the pairs (isolated word, point on it); the triples (two isolated words, common point), where two isolated words have exactly 5 common points because their δ is 0; the points of c_k; the pairs (isolated word, common point with c_k), again 5 for each isolated word; and the common points of c_k and c_l. A code thus gives a solution in nonnegative integers; to exclude G it suffices that there is no solution in nonnegative real numbers.

4 Pairs of points

In this section $S > 0$. Let $j \ne j'$ be points of types $i = (b, a)$ and $i' = (b', a')$, and let t be the number of isolated words that contain both (their *overlap*). Then $\max(0, a + a' - u) \le t \le \min(a, a')$. On the isolated words, z_j and $z_{j'}$ differ in $a + a' - 2t$ places. So, with

$$e_i = 1 - \beta^{\mathsf{T}} z_j = 1 - \beta_v^{\mathsf{T}} z_b - \tfrac{3}{16}(u - 2a),$$

the entries of P are

$$P_{jj'} = p_{ii'}(t) := z_b^{\mathsf{T}} H_v^{-1} z_{b'} + \frac{u - 2a - 2a' + 4t}{32} + \frac{e_i e_{i'}}{S}, \qquad P_{jj} = p_i := z_b^{\mathsf{T}} H_v^{-1} z_b + \frac{u}{32} + \frac{e_i^2}{S}.$$

By Lemma 3(ii) every 2×2 principal minor of $I - P$ is nonnegative:

$$p_{ii'}(t)^2 \le (1 - p_i)(1 - p_{i'}). \tag{7}$$

A triple (i, i', t) of two admissible types and an overlap is called *admissible* if it satisfies (7).

For an admissible triple let $Z_{i,i',t}$ be the number of ordered pairs (j, j') of distinct points with types i, i' and overlap t. Then $Z_{i,i',t} = Z_{i',i,t} \ge 0$. Each of them counts ordered pairs, so no factor 2 arises when the two are treated as one unknown.

Fix a point j of type $i = (b, a)$. For each of the other 31 points j', of type (b', a') and with overlap t, put $h = a' - t$; this is the number of isolated words that contain j' but not j. Adding over the 31 points j' we find:

Quantity added over j'	Sum	Reason
1	31	
t	$12a$	an isolated word on j has 12 other points
h	$13(u-a)$	an isolated word not on j has 13 points
$\binom{t}{2}$	$4\binom{a}{2}$	two isolated words on j have 4 other common points
th	$5a(u-a)$	isolated words on j and not on j have 5 common points
$\binom{h}{2}$	$5\binom{u-a}{2}$	two isolated words not on j have 5 common points
b'_k	$13-b_k$	c_k has $13-b_k$ points other than j
$t\,b'_k$	$a(5-b_k)$	c_k and an isolated word on j
$h\,b'_k$	$5(u-a)$	c_k and an isolated word not on j
$b'_k b'_l$	$5-\delta_{kl}-b_k b_l$	c_k and c_l, for $k<l$

Add each of these identities over the N_i points j of type i. For every admissible type i and every line (φ,ρ) of the table this gives

$$\sum_{i',t} \varphi \cdot Z_{i,i',t} = \rho \cdot N_i, \tag{8}$$

where the sum is over the admissible triples with first type i.

5 The computation

For each of the 102 graphs G the following is done in exact rational or integer arithmetic.

Step 1. Compute H_v^{-1}, β and S. If $S<0$, then G is excluded by Lemma 3(i). This happens for 72 graphs.

Step 2. Otherwise, list the admissible types and consider the system (E1)–(E6) in unknowns $N_i \geq 0$. For 22 graphs it has no solution.

Step 3. For the remaining 8 graphs, which all have $S>0$, list the admissible triples and consider the system (E1)–(E6), (8) in unknowns $N_i \geq 0$ and $Z_{i,i',t} \geq 0$, with one unknown for $Z_{i,i',t}$ and $Z_{i',i,t}$. In all 8 cases it has no solution.

That a system $Ax=c$, $x\geq 0$ has no solution is shown as follows. The data contain an integer vector y with $A^{\mathsf{T}}y \leq 0$ in every coordinate and $c^{\mathsf{T}}y>0$. A solution $x\geq 0$ would give $0<c^{\mathsf{T}}y = x^{\mathsf{T}}A^{\mathsf{T}}y \leq 0$, see e.g. [2, Section 5.8]. The program builds A and c from the formulas of Sections 1–4 and checks these two properties of y in integer arithmetic.

The completeness of the list of 102 graphs is checked as follows. For $D=0,\ldots,5$ the program runs through all partitions of the $2D$ endpoints of D labelled edges into vertices such that no edge becomes a loop (there are $1,1,7,87,1657,43833$ of them), and verifies that each resulting multigraph is isomorphic to one in the list. Every multigraph with D edges and no isolated vertex arises in this way: label its edges and their endpoints, and put the endpoints at the same vertex into one part.

So no graph is the defect graph of a code with 27 words, and the Theorem follows. For the 8 graphs of Step 3 the sizes are as follows; the graphs are numbered as in the data files, and the last two columns give the value of $c^{\mathsf{T}}y$ in the two sets of data described in Section 7.

Graph	Admissible types	Unknowns Z	$c^{\mathsf{T}}y$, first set	$c^{\mathsf{T}}y$, second set
0	3	11	380100	15900
1	11	88	10166280956550	101015500
2	36	775	17332892502687	172741660
3	8	56	17332892538832	172678000
4	16	198	17332892475776	172725640
10	13	114	24491570217444	244335340
11	25	437	27158486349294	271010145
12	25	432	24491570206131	244342655

6 An example: the graph without edges

Here $D = 0$, $v = 0$, $u = 27$, $H = 32\,I$, $\beta = \frac{3}{16}\mathbf{1}$ and $S = 32 - 36 \cdot \frac{27}{32} = \frac{13}{8}$. A type is a number a, the number of words on the point, and $e_a = (6a - 65)/16$. The condition $P_{jj} \le 1$ reads $\frac{27}{32} + \frac{(6a-65)^2}{416} \le 1$, that is, $(6a - 65)^2 \le 65$, so $a \in \{10, 11, 12\}$; these values also satisfy (4). The equations (E1)–(E3) are

$$N_{10} + N_{11} + N_{12} = 32, \quad 10N_{10} + 11N_{11} + 12N_{12} = 351, \quad 45N_{10} + 55N_{11} + 66N_{12} = 1755,$$

with the unique solution $N_{10} = 6$, $N_{11} = 21$, $N_{12} = 5$. Evaluating (7) shows that the overlap of two points with a and a' words on them is restricted as follows.

(a, a')	$(10, 10)$	$(10, 11)$	$(10, 12)$	$(11, 11)$	$(11, 12)$	$(12, 12)$
overlap t	$2, 3$	$3, 4$	5	$3, 4, 5$	$4, 5$	4

Fix a point j with 12 words on it. Among the other 31 points, six have 10 words on them, 21 have 11, and four have 12; their overlaps with j are 5; 4 or 5; and 4. By the second line of the table in Section 4 the overlaps add up to $12 \cdot 12 = 144$. So exactly 14 of the 21 points have overlap 5. Then $\sum \binom{t}{2} = 20\binom{5}{2} + 11\binom{4}{2} = 266$, but by the fourth line of that table this sum is $4\binom{12}{2} = 264$. So the graph without edges is excluded.

7 Files

The repository `github.com/wille614/cw-code-upper-bounds` contains the list of graphs and, for every graph, the data used in Section 5, in two independent sets. Each set has its own checking program, with different implementations of the exact matrix inverse and of the isomorphism test.

First set:	`certificates/A32_16_13_first.json`,	`checks/projection_first.py`
Second set:	`certificates/A32_16_13_second.json`,	`checks/projection_second.py`

The command `python verify.py 32 16 13` runs both.

In the first set the types are ordered by b read as a binary number (b_k being bit k), then by a; in the second set b is ordered lexicographically, then a. In both, the pairs $k < l$ are in lexicographic order, and the equations are (E1)–(E6) in this order, followed, for each admissible type i, by the equations (8) in the order of the table of Section 4, written as $\sum \varphi \cdot Z - \rho \cdot N_i = 0$.

Remark. Bounds from the integrality of column weights, as in Lemma 1, are used in [1, Theorem 17].

References

[1] M. R. Best, A. E. Brouwer, F. J. MacWilliams, A. M. Odlyzko and N. J. A. Sloane, Bounds for binary codes of length less than 25, *IEEE Trans. Inform. Theory* **24** (1978) 81–93.

[2] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004.

[3] A. E. Brouwer, Bounds for binary constant weight codes, `https://aeb.win.tue.nl/codes/Andw.html`.